

CAMPUS HUB

ONE CAMPUS, INFINITE POSSIBILITIES

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INTRODUCTION



What is Campus Hub ?

- + Campus Hub is an Education Management System (EMS) designed to centralize student data, attendance, fee management, and more.
- + It provides an all-in-one platform for educational institutions to efficiently manage administrative tasks.
- + The system enhances communication between students, teachers, and administrators by providing a centralized platform for announcements, updates, and notifications.



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PROBLEM STATEMENT



Problem Statement

A need for a single, integrated platform that centralizes and secures student and educational data.

Challenges Faced

Inefficiency

Manual tracking of student data, attendance, and fees often leads to human errors.

Lack of Centralized Data

Educational institutions struggle with disconnected systems for managing student records and other administrative tasks.

Security Concerns

Sensitive student data was often not secure or properly managed.

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Why This Project?



What was the need for this project ?



Need for Efficiency

- Existing systems were prone to errors and lacked automation.
- Administrative processes, such as fee tracking and attendance, required manual effort and were time-consuming.



Challenges

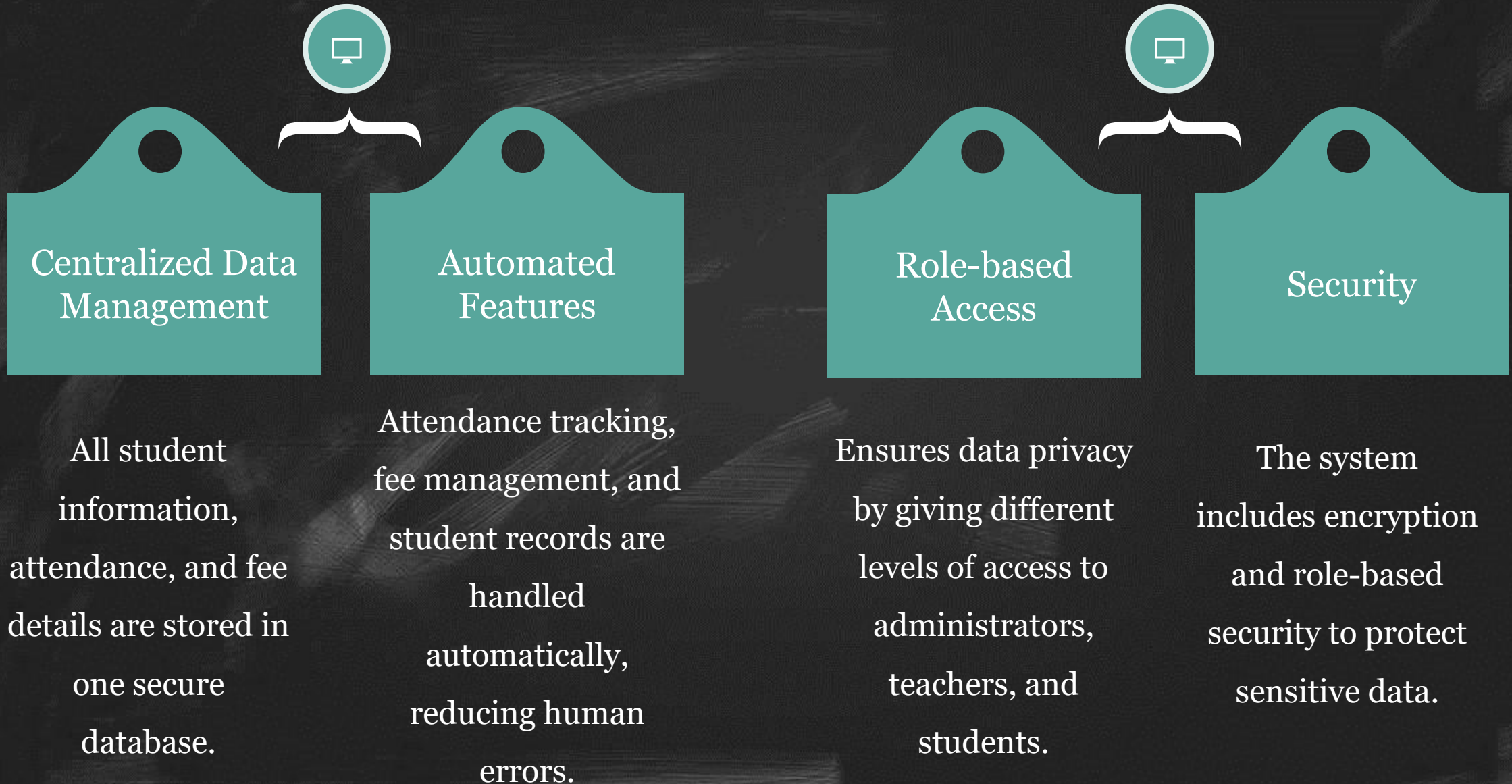
- Teachers, students, and administrators faced difficulties accessing and managing records.
- Data was scattered across different platforms, causing delays in decision-making and reporting.

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How it Solves the Problem ?



How “ Campus Hub ” solves the problem ?



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Technologies Used



Which Technologies are used ?



Java Technology

- Used for backend development.
- Ensures the system is strong, stable, and can grow with more users.



Client-Server Architecture

- Client: The interface where users (students, teachers, admins) interact.
- Server: Handles data and processes the logic.



MVC Design Pattern

- Separates the user interface, business logic, and data for better organization and easier management.

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Functional Requirements



Following are the Functional Requirements :



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Non-Functional Requirements



Following are the Non-Functional Requirements :

1

Performance

The system should handle at least 50 concurrent users without significant performance issues.

2

Usability

The system should be user-friendly and require minimal training.

3

Scalability

The system must be scalable to accommodate growing data and users.

4

Security

- All sensitive data must be encrypted.
- Role-based access ensures confidentiality.

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Reliability

- The system should have 99.9% uptime
- Automated daily backups ensure no data loss.

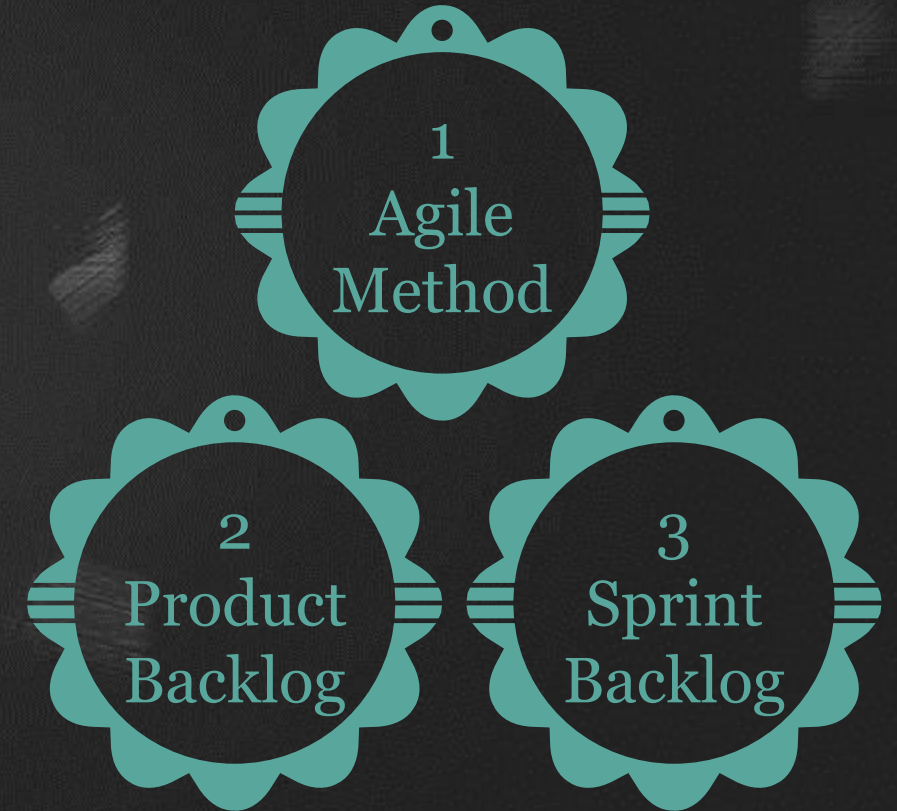
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Artifacts Overview



Backlog and Project Planning

- ① The system was developed using Agile Scrum, with iterative development and regular stakeholder feedback.
- ② A prioritized list of all system features, including student record management, attendance tracking, and fee management.
- ③ The team selects tasks for each sprint to ensure that high-priority features are developed first.



SRS(Software Requirement Specification)

- It defines what the system will do, how it will perform, and the constraints within which it must operate.



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The document serves as a blueprint for the system, helping developers and stakeholders understand the scope of the project and ensuring the system meets all user needs.

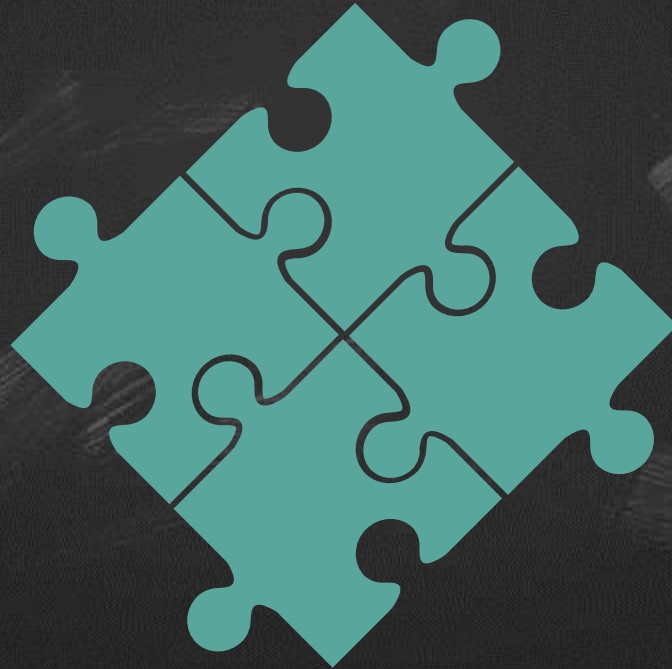
Diagrams

Context Diagram

Sequence Diagram

Use Case Diagram

Class Diagram



Architecture



Architecture

The client interacts with the system through the user interface (UI), while the server processes requests, handles business logic, and communicates with the database.



Design Pattern

The Model handles data logic, the View is responsible for the UI, and the Controller connects the two, ensuring that data flows properly between the system and the user interface.

Test Cases:



It validate the system's
functionality.



It ensure that key features
work as expected.

“ For example :

Test cases check whether the system correctly handles adding new student records, updating attendance, and managing fee status. These tests ensure the system operates smoothly, meeting all functional requirements.

”

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Future Enhancements



To further improve the “Campus Hub” system and adapt to evolving educational needs, the following enhancements are planned for the future.



Scalability for Larger Institutions :



- The system will be scaled to handle even more users and data as the institution grows

Mobile App Integration



- A mobile version of the Campus Hub for easy access by students and teachers on the go.

Advanced Reporting :



- Adding advanced reporting features for admins to monitor student performance, attendance, & fees.

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Conclusion



CONCLUSION

Summary : Campus Hub is an efficient solution for managing educational data, streamlining administrative tasks, and ensuring data security.

Impact: The system significantly reduces manual work, minimizes errors, and improves decision-making in educational institutions.



Acknowledgments : Thanks to the team members for their contributions in making this project a success.

THANK YOU